

Heritage of India: Advanced Monuments Classification using Artificial Intelligence

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Abstract—Tourism is India's most lucrative service industry. Many historical buildings and cultural practices originated in India. Every year, millions of people go to tourist hotspots, and millions more share photos they have taken there on social media and mobile devices with friends and family. The promotion and dissemination of cultural assets as well as tourism activities and landmarks in urban settings might benefit greatly from the use of artificial intelligence methods and machine learning algorithms, which are already being used in a wide variety of settings. Automatic identification and categorization of landmarks and tourist attractions might be an intriguing use case for these methods in the cultural and touristic sector. This study makes an effort to categorize India's historic sites. Hotels, travel companies, restaurants, government organizations, etc. all benefit from the classification of monuments so that they may better plan for the future and provide facilities that are up to date with visitor expectations. Using a deep learning technique, the suggested approach aims to retrieve characteristics and categorize the Indian landmarks that visitors visited.

Keywords— Machine learning, Accuracy, Classification, Heritage, Indian Monuments, Convolutional Neural Network, Deep Learning

I. INTRODUCTION

India is a stunning country, renowned for its vibrant culture and rich history. India has a long and illustrious past, and its great structures like the Taj Mahal serve as sturdy testaments to that fact. The historical monuments of India are a powerful testimony to the country's rich past and illustrious architecture. The Rajputana, Dravidian, and Mughal rulers who ruled India at the time of its construction were responsible for the bulk of India's iconic architecture[1]. India's historic architecture is a living testimony to the power and creativity of India's emperors and craftsmen. India's tourism industry has boomed recently, with more and more visitors from all over the globe drawn to the country by its stunning natural scenery and the government's ongoing attempts to preserve its historic landmarks[2].

Due to the widespread availability of relatively inexpensive electronic devices like cameras, mobiles, and Handy cams, a vast quantity of multimedia data in the shape of photos, music, and video has been created over many years. Family and friends who could not be there in person could still feel like they were a part of the action by seeing the photos, videos, and audio recordings that were sent to them over the internet or email. With millions of photos posted every day, manually sorting them into albums based

on which landmarks were visited would be an impossible task.

Many other types of tourism, such as those centered on medical care, business, and competition, are included in the travel and tourism sector. The principal objective of this industry is to develop and promote tourism so as to guarantee job generation and economic prosperity by maintaining India's competitiveness as a tourist destination and expanding and improving current tourism offerings[3][4]. The government is also devoting significant resources to marketing and programmes geared at visitors, and it is carefully maintaining these sites so that they may be appreciated by future generations. Large numbers of international guests and delegates go to India annually. Statistics on tourism in India for the year 2021–22 show that there were 3,18,673 international tourists, down from 4,15,859 the previous year. ii. There was a 98 percent annual increase in domestic tourism to the United States in 2021–22, with 2,60,46,891 tourists as opposed to 1,31,53,076 in the previous year.[5].

A. Machine Learning

Machine learning (ML) is a method through which software may improve its prediction skills over time without being explicitly programmed to do so. Many modern predictive models use machine learning algorithms, which analyze past data to make predictions [6], [7].

An important way to classify classical machine learning is by the means through which a predictive algorithm improves its performance. Supervised, unsupervised, semi-supervised, and reinforcement learning are the four cornerstones. Data scientists utilize many algorithms for prediction purposes, and the one they choose is determined by the data they are working with[8].

Supervised learning: Data scientists categorize training data and discover factors for correlations. Input and output are specified.

Unsupervised learning: This subfield of machine learning utilizes unlabeled data to educate its algorithms. Data from every source is checked for duplication. The supplied data and the conclusions or suggestions are predetermined. Semi-supervised learning is a more recent machine learning technique that combines aspects of the previous two types of ML. It is up to the model to find and understand the dataset on its own, even if data scientists provide it with a significant quantity of labelled training data.

Reinforcement learning: Algorithms are trained by data scientists who give them instructions on how to do a job

and then provide feedback in the form of positive or negative reinforcement. Most of the time, however, the algorithm is left to its own devices to figure out what to do next.

Deep Learning: It entails a network of neurons with three or more layers. These neural networks "learn" from massive volumes of data by mimicking the activity of the human brain, however, they are still far from matching its capabilities. One layer of a neural network can still produce predictions with some degree of accuracy, but more advanced networks with hidden layers are better able to improve and enhance their predictions. In contrast to traditional machine learning, deep learning does not need as much preparation of data. By ingesting and processing unstructured data like text and pictures, these algorithms reduce reliance on human specialists by automating feature extraction[9].

In order to categorize India's historical sites, researchers have turned to supervised learning and deep learning frameworks. Here is the paper's outline: In Section II, we present a thorough classification of ML based on its useful qualities and a critical evaluation of previous work. In Part III, we break down the procedure and its essential elements. Test repercussions are mentioned in Section IV. In the last section, we provide some final thoughts.

II. LITERATURE REVIEW

Bhatt M.S. et al. [10] supplied a Generalized Co-occurrence Matrix (GCM) with 64 gray levels and distance data to a Genetic Programming (GP) System (3, 6, 9, 12, and 15). Seven functions and two operators produce the spatial descriptor that 15 GCO matrices employ in a GP algorithm. Each GCM of size 64x64 is taken into consideration. The genetic programming-generated spatial descriptor is tested against a database of manually-created Indian monuments, including the Taj Mahal, Qutub Minar, Golden Temple, and India Gate. GP uses a 10-fold cross-validated linear SVM fitness function. 92% Success rate was achieved.

Two-dimensional Estimation of Signal Parameters using Rotationally Invariant Techniques (ESPRIT) and the Gabor Filter was the basis for a proposal for content-based picture retrieval by Youness et al. [11]. For this goal, they employed the Brodatz grayscale picture dataset, which has 13 texture classes and 16 samples for each class. As a group, they were accurate 80.19 percent of the time.

Content-based picture retrieval was used by Nazarloo et al. [12] to determine gender. One of the most significant human biometrics, the face may provide a wealth of information about a person. They combined facial features based on Gabor filters and Local Binary Patterns to determine sex. An impressive 92.5 percent accuracy was attained in the categorization using a Self-Organized Map.

For content-based picture identification using niblack's thresholding, Das et al. [13], [14] emphasized the significance of categorized queries. Using the thresholding provided by niblack, they independently performed binarization to the Red, Blue, and Green channels. Higher and lower averages as well as upper and lower standard deviations are computed from the binarized pictures. A feature vector of size 12 is produced by extracting four features from each plane. They put the software through its paces using the 'Wang Database' and 'OT-Scene' as test cases. For the "Wang Database," the

best results were obtained using an ANN with a MLP; for the "OT-Scene" database, the best results were obtained using a Support Vector Machine.

Desai et al. [15] stressed the significance of monument categorization for assessing archaeological discoveries and categorizing monuments. The visual features of artifacts are highlighted in museums and art galleries. To address this problem, we built a CBIR platform on top of visual shape-based features and texture features. Correlation in the Grayscale matrix method was used to isolate textural characteristics, whereas morphological techniques were performed to extract shapes. Performance is tested between the Canny and Sobel edge detection methods using a dataset consisting of 500 photos from 5 distinct classes.

III. MATERIALS AND METHODS

A. Data Set

Indian monuments image dataset [16] is used in this study. The collection has 24 categories, with around 200 photos in each category. In all, there are 4895 pictures in the collection, all of which are views of various Indian monuments from various vantage points.

The dataset sample images of some famous monuments and heritage places in India are shown in Fig. 1.

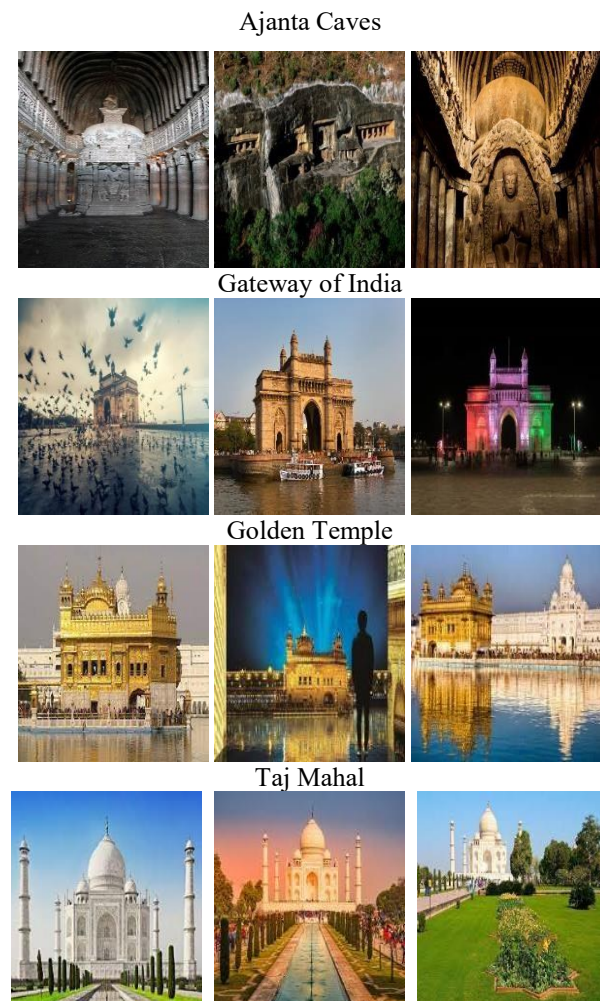


Fig. 1. Sample dataset images

B. Data Augmentation

Data Augmentation is used at this stage to enhance the pictures before they are used. This is done so that less of a "perception gap" exists between the photographer and the subject while photographing historic sites. Identical structure to the monument. Various photographs may be captured by taking them at varied times of the day or under different lighting conditions. Through the use of data augmentation, the system's efficiency may be raised by creating many instances of the provided picture[17].

Image data from the whole dataset is flipped, rotated, translated, and zoomed at random. The functions for preparing images are available in the TensorFlow image module.

C. Methodology

In this study, we use a stack of machine learning techniques, including Naive Bayes, Support Vector Machine, and K-nearest neighbour, on an enhanced data set. Together with transfer learning techniques, we use Convolutional Neural Network and models such as MobileNet[18], InceptionNet[19], SqueezeNet[20], VGG16, and VGG19. The illustration of the suggested structure is depicted in fig. 2.

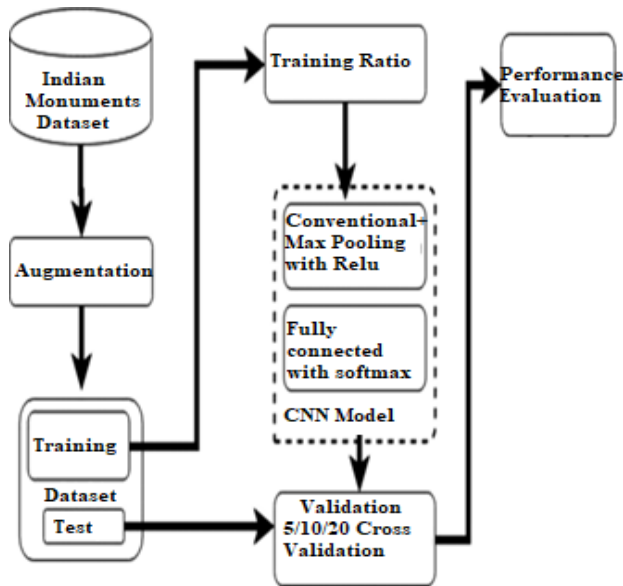


Fig. 2. Mechanisms of the Proposed Model's Actualization

Throughout each trial, we used the Convolutional Neural Network (CNN) and Transfer Learning architecture provided by the Keras Python module. Training data accounted for 70% of the dataset, while validation data made up the remaining 30%. Before being fed into the deep architectures, both the training and validation datasets were imported, supplemented, and normalised as Batch Dataset objects in Tensor Flow v2.4.0.

IV. RESULTS

A. Experimental Setup

The research used a Core i7 processor, GeForce MX150 NVIDIA graphics card, 8 GB DDRAM, and a 500 GB SSD. Anaconda, a Python IDE, version 3.9.7, was utilised.

B. Results

The performance of machine learning models has traditionally been measured using four metrics: Classification accuracy(CA), F1-score(F1), recall(R), and precision(P).

1) Fundamental Machine Learning Algorithms

Table I and Figure 3 show the results of the most fundamental ML models.

TABLE I. PERFORMANCE OF BASE MACHINE LEARNING MODELS

Model	CA	F1	P	R
Naive Bayes	0.688	0.698	0.728	0.688
kNN	0.844	0.844	0.852	0.844
Stack	0.894	0.894	0.897	0.894
SVM	0.904	0.904	0.906	0.904

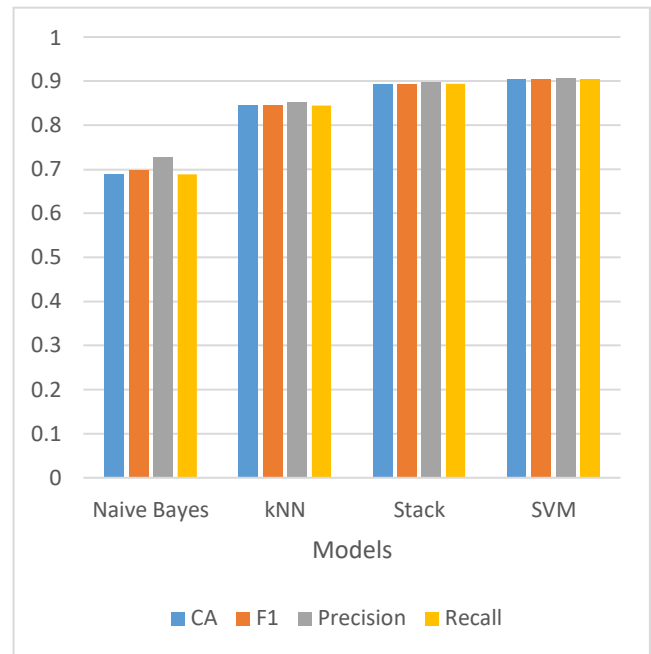


Fig. 3. Performance Evaluation

2) Convolutional Neural Network

This paper's goal was met by using a deep neural network method to the same datasets in order to build and evaluate a classification model. The outcomes of applying CNN to the expanded dataset are shown in tables II. The validation accuracy for CNN using the training dataset was 92.24%.

TABLE II. CNN PERFORMANCE

Model	Number of folds	CA	F1	P	R
CNN	20	0.922465	0.92186	0.922855	0.922465

For each epoch training and validation accuracy is depicted in fig. 4.

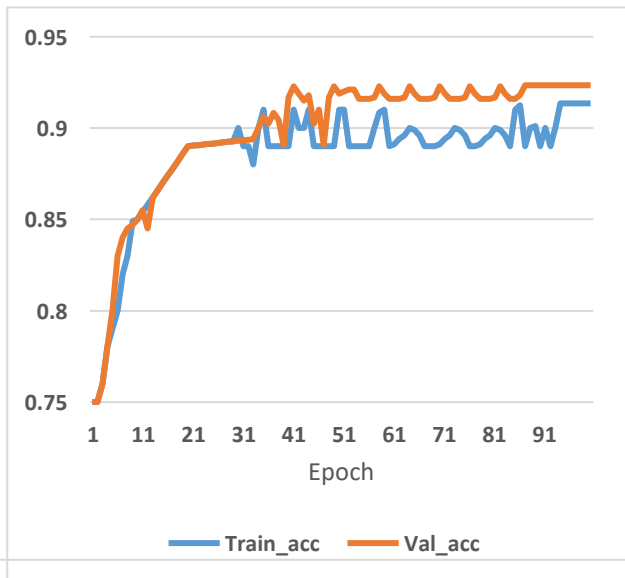


Fig. 4. CNN: Train accuracy Vs Validation accuracy

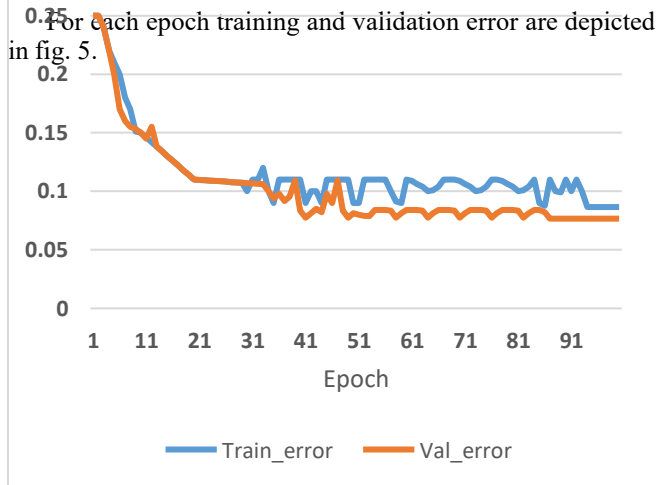


Fig. 5. CNN: Train accuracy Vs Validation accuracy

3) Deep Learning Architectures

The outcomes of applying deep learning models to the augmented dataset with 5, 10, and 20 cross-validation are shown in Tables III.

TABLE III. PERFORMANCE OF DEEP LEARNING ARCHITECTURE

Model	Number of folds	CA	F1	P	R
Mobile Net	5	0.851	0.845	0.834	0.85
VGG16		0.871	0.864	0.871	0.871
VGG19		0.9	0.91	0.9	0.9

RestNet		0.912	0.912	0.91	0.91
SqueezeNet		0.677	0.687	0.677	0.687
InceptionNet		0.90222	0.90283	0.90383	0.90282
Mobile Net		0.89	0.891	0.891	0.89
VGG16	10	0.886	0.886	0.88	0.88
VGG19		0.91	0.912	0.91	0.91
RestNet		0.911	0.911	0.91	0.91
SqueezeNet		0.67	0.68	0.67	0.68
InceptionNet	20	0.90222	0.90283	0.90383	0.90282
Mobile Net		0.89	0.89	0.89	0.89
VGG16		0.887	0.887	0.88	0.88
VGG19		0.9	0.91	0.9	0.9
RestNet	20	0.91	0.91	0.9	0.9
SqueezeNet		0.67	0.68	0.67	0.68
InceptionNet		0.90222	0.90283	0.90383	0.90282

4) Comparison

Typical ML models, deep learning architectures, and CNN were evaluated for their validation accuracy, training accuracy, precision, and recall on augmented datasets, and the results are shown in Fig. 6. Fig. 6 clearly demonstrates that the Convolutional Neural Network provides the best results.

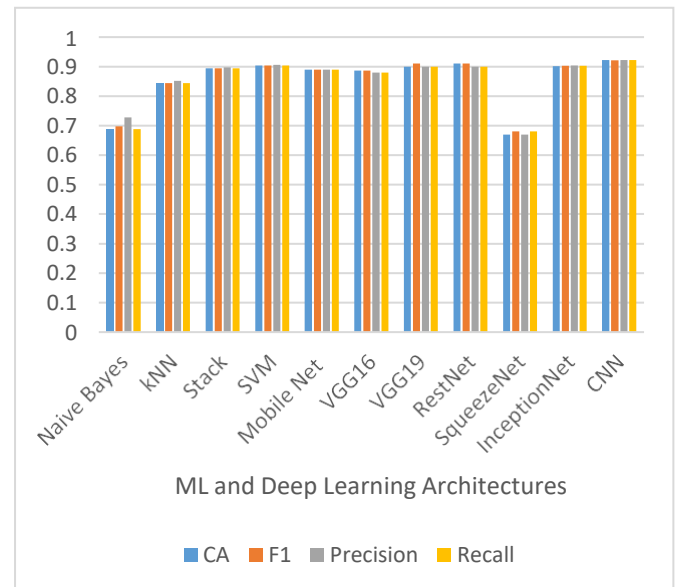


Fig. 6. Performance evaluation

V. CONCLUSION

As the number of tourists visits a certain area, there is a corresponding rise in curiosity to learn more about the landmarks and history behind them. Simply said, Automatic Monument Recognition is the process of categorizing and tagging Monuments. Due to the fact that monuments are three- dimensional constructions, the photos of these structures shot from various perspectives will always seem somewhat different. Data Augmentation techniques like as random flipping, rotating, translating, zooming, contrasting, tinting, and saturating were utilized to lessen the issue of divergent interpretations. The initial step in our automated identification of Indian cultural monuments is to use classical machine learning methods, followed by Convolutional Neural Networks and deep learning architectures. The feature extractor and transfer trained network improved classification accuracy above standard machine learning approaches. Through extensive testing, we have shown that the Convolutional Neural Network significantly outperforms other common machine-learning approaches. CNN has a validation accuracy of 92.24%.

Applying appropriate pre-processing methods to the dataset, with an emphasis on reducing noise and extraneous data, may further enhance performance. Overfitting is a problem with deep learning architectures. A number of techniques, including dropout and cross-validation, may be used to address this issue.

REFERENCES

- [1] E. Vanina, "Hindus and Muslims in Medieval North India: Stages of Reciprocal Perception," *Studies in Peoples History*, vol. 8, no. 1, pp. 47–61, Jun. 2021, doi: 10.1177/2348448921997031.
- [2] V. Kaushal and S. Srivastava, "Hospitality and tourism industry amid COVID-19 pandemic: Perspectives on challenges and learnings from India," *Int J Hosp Manag*, vol. 92, p. 102707, Jan. 2021, doi: 10.1016/j.ijhm.2020.102707.
- [3] R. S. Godara and D. J. Fetrat, "Impact of tourism industry on the Indian economy: a multidimensional analysis," *International Journal of Economic Policy in Emerging Economies*, vol. 15, no. 2–4, pp. 351–367, 2022, doi: 10.1504/IJEP.2022.121352.
- [4] S. Khan and S. M. Freeda Maria, "What innovations would enable the tourism and hospitality industry in India to re-build?," *Worldwide Hospitality and Tourism Themes*, vol. ahead-of-print, no. ahead-of-print, 2022, doi: 10.1108/WHATT-05-2022-0053/FULL/XML.
- [5] "India Tourism Statistics 2022 | Ministry Of Tourism | Government of India." <https://tourism.gov.in/annual-reports/india-tourism-statistics-2022> (accessed Oct. 31, 2022).
- [6] R. G. Tiwari, A. K. Agarwal, R. K. Kaushal, and N. Kumar, "Prophetic Analysis of Bitcoin price using Machine Learning Approaches," in *Proceedings of IEEE International Conference on Signal Processing, Computing and Control*, 2021, vol. 2021-Octob. doi: 10.1109/ISPCC53510.2021.9609419.
- [7] R. G. Tiwari, M. Husain, B. Gupta, and A. Agrawal, "Amalgamating contextual information into recommender system," in *Proceedings - 3rd International Conference on Emerging Trends in Engineering and Technology, ICETET 2010*, 2010. doi: 10.1109/ICETET.2010.110.
- [8] R. G. Tiwari, M. Husain, V. Srivastava, and A. Agrawal, "Web personalization by assimilating usage data and semantics expressed in ontology terms," in *International Conference and Workshop on Emerging Trends in Technology 2011, ICWET 2011 - Conference Proceedings*, 2011. doi: 10.1145/1980022.1980133.
- [9] A. K. Agarwal, R. G. Tiwari, V. Khullar, and R. K. Kaushal, "Transfer Learning Inspired Fish Species Classification," in *Proceedings of the 8th International Conference on Signal Processing and Integrated Networks, SPIN 2021*, 2021. doi: 10.1109/SPIN52536.2021.9566067.
- [10] M. S. Bhatt and T. P. Patalia, "Genetic programming evolved spatial descriptor for Indian monuments classification," *2015 IEEE International Conference on Computer Graphics, Vision and Information Security, CGVIS 2015*, pp. 131–136, Apr. 2016, doi: 10.1109/CGVIS.2015.7449908.
- [11] C. Youness, E. A. Khalid, O. Mohammed, and A. Brahim, "New Method of Content Based Image Retrieval based on 2-D ESPRIT Method and the Gabor Filters," *TELKOMNIKA Indonesian Journal of Electrical Engineering*, vol. 15, no. 2, pp. 313–320, Aug. 2015, doi: 10.11591/TJEE.V15I2.1544.
- [12] E. Parcham, M. Nazarloo, and R. A. Pourani, "Gender Classification Using Hybrid of Gabor Filters and Binary Features of an Image I Work in 'license plate recognition' and 'object detection' with deep learning logarithm View project Gender Classification Using Hybrid of Gabor Filters and Binary Features of an Image," *International Journal of Electrical and Computer Engineering (IJECE)*, vol. 4, no. 4, pp. 539–547, 2014.
- [13] DasRik, ThepadeSudeep, BhattacharyaSubhajit, and GhoshSaurav, "Retrieval architecture with classified query for content based image recognition," *Applied Computational Intelligence and Soft Computing*, vol. 2016, pp. 1–9, Jan. 2016, doi: 10.1155/2016/1861247.
- [14] S. Thepade, R. Das, and S. Ghosh, "Novel technique in block truncation coding based feature extraction for content based image identification," *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 9030, pp. 55–76, 2015, doi: 10.1007/978-3-662-47074-9_4/COVER.
- [15] P. Desai, J. Pujari, N. H. Ayachit, and V. K. Prasad, "Classification of archaeological monuments for different art forms with an application to CBIR," *Proceedings of the 2013 International Conference on Advances in Computing, Communications and Informatics, ICACCI 2013*, pp. 1108–1112, 2013, doi: 10.1109/ICACCI.2013.6637332.
- [16] "Indian monuments image dataset | Kaggle." <https://www.kaggle.com/datasets/danushkumarv/indian-monuments-image-dataset> (accessed Oct. 31, 2022).
- [17] C. Shorten and T. M. Khoshgoftaar, "A survey on Image Data Augmentation for Deep Learning," *J Big Data*, vol. 6, no. 1, pp. 1–48, Dec. 2019, doi: 10.1186/S40537-019-0197-0/FIGURES/33.
- [18] H. Y. Chen and C. Y. Su, "An Enhanced Hybrid MobileNet," *2018 9th International Conference on Awareness Science and Technology, iCAST 2018*, pp. 308–312, Oct. 2018, doi: 10.1109/ICAWST.2018.8517177.
- [19] Shukla, P.K., Sandhu, J.K., Ahirwar, A., Ghai, D., Maheshwary, P. and Shukla, P.K., 2021. Multiobjective genetic algorithm and convolutional neural network based COVID-19 identification in chest X-ray images. *Mathematical Problems in Engineering*, 2021, pp.1-9.
- [20] Koundal, D., Gupta, S. and Singh, S., 2018. Computer aided thyroid nodule detection system using medical ultrasound images. *Biomedical Signal Processing and Control*, 40, pp.117-130.